

AI-POWERED SKILL GAP ANALYSIS REPORT

Candidate: Arjun Sharma

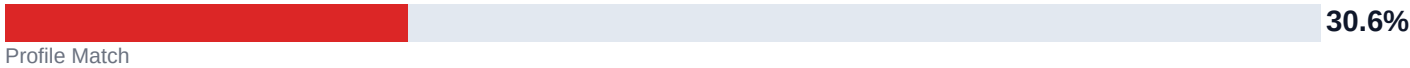
Target Role: Machine Learning Engineer

Date: 07 May 2026, 06:12 AM

JDs Analysed: 10

Executive Summary

Your profile shows a early-stage match (30.6%) for Machine Learning Engineer roles. You matched 15 of the required skills. Analysis of 10 live job descriptions revealed 59 skill gaps — 2 critical, 12 moderate, and 45 minor. This role requires significant upskilling. Follow the roadmap systematically. Estimated time to close all gaps: 45 weeks (~15 hrs/week).



15
Skills Matched

2
Critical Gaps

12
Moderate Gaps

45
Minor Gaps

45w
Est. Timeline

Matched Skills

BigQuery · Docker · ETL · Feature Engineering · Git · Linux · Machine Learning · Model Deployment · PostgreSQL · Python · REST API · Redis · SQL · Scikit-learn · Statistical Analysis

Skill Gap Analysis

Skill	Category	Priority	JD Freq.	Why It Matters
Kubernetes	Cloud & DevOps	CRITICAL	7	Required in 70% of 10 JDs analysed — core skill for this role.
AWS	Cloud & DevOps	CRITICAL	6	Appears in 60% of JDs — strongly recommended for this role.
Apache Kafka	Data Engineering	MODERATE	5	Appears in 50% of JDs — strongly recommended for this role.
CI/CD	Cloud & DevOps	MODERATE	4	Appears in 40% of JDs — strongly recommended for this role.
System Design	Tools & Practices	MODERATE	4	Appears in 40% of JDs — strongly recommended for this role.
Deep Learning	Data Science & Machine Learning	MODERATE	3	Mentioned in 30% of JDs — adds significant competitive advantage.
LLM	Data Science & Machine Learning	MODERATE	3	Mentioned in 30% of JDs — adds significant competitive advantage.

Skill	Category	Priority	JD Freq.	Why It Matters
Apache Spark	Data Engineering	MODERATE	3	Mentioned in 30% of JDs — adds significant competitive advantage.
Data Structures	Tools & Practices	MODERATE	3	Mentioned in 30% of JDs — adds significant competitive advantage.
Prompt Engineering	AI & Emerging Tech	MODERATE	3	Mentioned in 30% of JDs — adds significant competitive advantage.
Apache Airflow	Data Engineering	MODERATE	3	Mentioned in 30% of JDs — adds significant competitive advantage.
Generative AI	AI & Emerging Tech	MODERATE	3	Mentioned in 30% of JDs — adds significant competitive advantage.
RAG	Data Science & Machine Learning	MODERATE	3	Mentioned in 30% of JDs — adds significant competitive advantage.
LangChain	Data Science & Machine Learning	MODERATE	3	Mentioned in 30% of JDs — adds significant competitive advantage.
Go	Programming Languages	MINOR	4	Appears in 40% of JDs — strongly recommended for this role.
MLflow	Data Science & Machine Learning	MINOR	3	Mentioned in 30% of JDs — adds significant competitive advantage.
Fine-tuning LLMs	AI & Emerging Tech	MINOR	3	Mentioned in 30% of JDs — adds significant competitive advantage.
FastAPI	Web Development	MINOR	3	Mentioned in 30% of JDs — adds significant competitive advantage.
TensorFlow	Data Science & Machine Learning	MINOR	2	Mentioned in 20% of JDs — adds significant competitive advantage.
PyTorch	Data Science & Machine Learning	MINOR	2	Mentioned in 20% of JDs — adds significant competitive advantage.
Data Warehousing	Data Engineering	MINOR	2	Mentioned in 20% of JDs — adds significant competitive advantage.
MLOps	AI & Emerging Tech	MINOR	2	Mentioned in 20% of JDs — adds significant competitive advantage.
Hugging Face	Data Science & Machine Learning	MINOR	2	Mentioned in 20% of JDs — adds significant competitive advantage.
Terraform	Cloud & DevOps	MINOR	2	Mentioned in 20% of JDs — adds significant competitive advantage.
Azure	Cloud & DevOps	MINOR	2	Mentioned in 20% of JDs — adds significant competitive advantage.
Communication	Soft Skills	MINOR	2	Mentioned in 20% of JDs — adds significant competitive advantage.
BERT	Data Science & Machine Learning	MINOR	2	Mentioned in 20% of JDs — adds significant competitive advantage.

Skill	Category	Priority	JD Freq.	Why It Matters
dbt	Data Engineering	MINOR	2	Mentioned in 20% of JDs — adds significant competitive advantage.
Vector Database	Databases	MINOR	2	Mentioned in 20% of JDs — adds significant competitive advantage.
Databricks	Data Engineering	MINOR	2	Mentioned in 20% of JDs — adds significant competitive advantage.
Site Reliability Engineering	Cloud & DevOps	MINOR	2	Mentioned in 20% of JDs — adds significant competitive advantage.
Natural Language Processing	Data Science & Machine Learning	MINOR	1	Nice-to-have skill that appears in 10% of JDs.
Ansible	Cloud & DevOps	MINOR	2	Mentioned in 20% of JDs — adds significant competitive advantage.
Prometheus	Cloud & DevOps	MINOR	2	Mentioned in 20% of JDs — adds significant competitive advantage.
Scala	Programming Languages	MINOR	2	Mentioned in 20% of JDs — adds significant competitive advantage.
LangGraph	AI & Emerging Tech	MINOR	2	Mentioned in 20% of JDs — adds significant competitive advantage.
Problem Solving	Soft Skills	MINOR	1	Nice-to-have skill that appears in 10% of JDs.
Microservices	Web Development	MINOR	1	Nice-to-have skill that appears in 10% of JDs.
Grafana	Cloud & DevOps	MINOR	2	Mentioned in 20% of JDs — adds significant competitive advantage.
Great Expectations	Data Engineering	MINOR	2	Mentioned in 20% of JDs — adds significant competitive advantage.
Django	Web Development	MINOR	1	Nice-to-have skill that appears in 10% of JDs.
Unit Testing	Tools & Practices	MINOR	1	Nice-to-have skill that appears in 10% of JDs.
Data-Driven Decision Making	Soft Skills	MINOR	1	Nice-to-have skill that appears in 10% of JDs.
Snowflake	Databases	MINOR	1	Nice-to-have skill that appears in 10% of JDs.
Shell Scripting	Programming Languages	MINOR	1	Nice-to-have skill that appears in 10% of JDs.
Project Management	Soft Skills	MINOR	1	Nice-to-have skill that appears in 10% of JDs.
Jenkins	Cloud & DevOps	MINOR	1	Nice-to-have skill that appears in 10% of JDs.
R	Programming Languages	MINOR	1	Nice-to-have skill that appears in 10% of JDs.
Elasticsearch	Databases	MINOR	1	Nice-to-have skill that appears in 10% of JDs.
Vector Embeddings	AI & Emerging Tech	MINOR	1	Nice-to-have skill that appears in 10% of JDs.

Skill	Category	Priority	JD Freq.	Why It Matters
GitHub Actions	Cloud & DevOps	MINOR	1	Nice-to-have skill that appears in 10% of JDs.
Design Patterns	Tools & Practices	MINOR	1	Nice-to-have skill that appears in 10% of JDs.
Leadership	Soft Skills	MINOR	1	Nice-to-have skill that appears in 10% of JDs.
spaCy	Data Science & Machine Learning	MINOR	1	Nice-to-have skill that appears in 10% of JDs.
LightGBM	Data Science & Machine Learning	MINOR	1	Nice-to-have skill that appears in 10% of JDs.
CrewAI	AI & Emerging Tech	MINOR	1	Nice-to-have skill that appears in 10% of JDs.
Flink	Data Engineering	MINOR	1	Nice-to-have skill that appears in 10% of JDs.
Recommendation Systems	Data Science & Machine Learning	MINOR	1	Nice-to-have skill that appears in 10% of JDs.
Nginx	Cloud & DevOps	MINOR	1	Nice-to-have skill that appears in 10% of JDs.

■ Learning Roadmap

Estimated completion: **45 weeks** at ~15 hrs/week

Skill	Course	Platform	Duration	Level	Free?
Kubernetes	Kubernetes for the Absolute Beginners	Udemy	6 hours	Beginner	Paid
Kubernetes	Kubernetes Official Documentation	kubernetes.io	Self-paced	Intermediate	FREE
AWS	AWS Certified Solutions Architect	Udemy	27 hours	Intermediate	Paid
AWS	AWS Cloud Practitioner Essentials	AWS	6 hours	Beginner	FREE
Apache Kafka	Apache Kafka Series — Beginner to Intermediate	Udemy	8 hours	Beginner	Paid
Apache Kafka	Kafka Official Documentation	Apache.org	Self-paced	Intermediate	FREE
CI/CD	DevOps Beginners to Advanced with Projects	Udemy	27 hours	Beginner	Paid
CI/CD	GitHub Actions Documentation	GitHub	Self-paced	Beginner	FREE
System Design	System Design Primer	GitHub	Self-paced	Intermediate	FREE
System Design	Grokking System Design Interview	Educative	20 hours	Intermediate	Paid
Deep Learning	Deep Learning Specialization	Coursera	5 months	Intermediate	Paid
Deep Learning	fast.ai Practical Deep Learning	fast.ai	8 weeks	Intermediate	FREE
Deep Learning	MIT 6.S191 Introduction to Deep Learning	YouTube	3 weeks	Intermediate	FREE
LLM	LLM Bootcamp — Full Stack Deep Learning	FSDL	6 hours	Intermediate	FREE
LLM	Building Systems with the ChatGPT API	DeepLearning.AI	4 hours	Beginner	FREE
Apache Spark	Apache Spark and Python for Big Data	Udemy	12 hours	Intermediate	Paid
Apache Spark	Databricks Academy Lakehouse Fundamentals	Databricks	5 hours	Beginner	FREE

Skill	Course	Platform	Duration	Level	Free?
Data Structures	Algorithms Specialization	Coursera	4 months	Intermediate	Paid
Data Structures	LeetCode Study Plans	LeetCode	Self-paced	Intermediate	FREE
Prompt Engineering	ChatGPT Prompt Engineering for Developers	DeepLearning.AI	3 hours	Beginner	FREE
Prompt Engineering	Learn Prompting	learnprompting.org	Self-paced	Beginner	FREE
Apache Airflow	The Complete Hands-On Introduction to Apache Airflow	Udemy	8 hours	Intermediate	Paid
Apache Airflow	Airflow Official Tutorial	Apache.org	Self-paced	Beginner	FREE
Generative AI	Generative AI for Everyone	Coursera	3 weeks	Beginner	Paid
RAG	Building and Evaluating Advanced RAG	DeepLearning.AI	3 hours	Intermediate	FREE
RAG	LangChain RAG in-depth	YouTube	3 hours	Intermediate	FREE
LangChain	LangChain for LLM Application Development	DeepLearning.AI	4 hours	Intermediate	FREE
LangChain	LangChain Crash Course	YouTube	2 hours	Beginner	FREE
Go	Go — Search on Coursera	Coursera	Varies	Beginner	Paid
MLflow	MLflow Official Documentation	MLflow.org	Self-paced	Intermediate	FREE
Fine-tuning LLMs	Finetuning Large Language Models	DeepLearning.AI	3 hours	Intermediate	FREE
Fine-tuning LLMs	QLoRA Fine-tuning Guide	Hugging Face	Self-paced	Advanced	FREE
FastAPI	FastAPI — Search on Coursera	Coursera	Varies	Beginner	Paid
TensorFlow	TensorFlow Developer Certificate	Coursera	4 months	Intermediate	Paid
TensorFlow	TensorFlow Official Tutorials	TensorFlow.org	Self-paced	Beginner	FREE
PyTorch	Official PyTorch Tutorials	PyTorch.org	Self-paced	Intermediate	FREE
PyTorch	PyTorch for Deep Learning Bootcamp	Udemy	17 hours	Intermediate	Paid
Data Warehousing	Data Warehouse Fundamentals	Udemy	9 hours	Beginner	Paid
MLOps	MLOps Zoomcamp	DataTalks.Club	9 weeks	Intermediate	FREE
MLOps	Machine Learning Engineering for Production (MLOps)	Coursera	4 months	Advanced	Paid
Hugging Face	Hugging Face NLP Course	Hugging Face	6 weeks	Intermediate	FREE
Terraform	Terraform for the Absolute Beginners	Udemy	5 hours	Beginner	Paid
Azure	Azure — Search on Coursera	Coursera	Varies	Beginner	Paid
Communication	Communication — Search on Coursera	Coursera	Varies	Beginner	Paid
BERT	BERT Research Paper Explained	YouTube	1 hour	Advanced	FREE
dbt	dbt Fundamentals	dbt Learn	5 hours	Beginner	FREE

Skill	Course	Platform	Duration	Level	Free?
Vector Database	Vector Databases from the Ground Up	DeepLearning.AI	3 hours	Beginner	FREE
Databricks	Databricks — Search on Coursera	Coursera	Varies	Beginner	Paid
Site Reliability Engineering	Site Reliability Engineering — Search on Coursera	Coursera	Varies	Beginner	Paid
Natural Language Processing	Hugging Face NLP Course	Hugging Face	6 weeks	Intermediate	FREE
Natural Language Processing	NLP Specialization	Coursera	4 months	Intermediate	Paid
Natural Language Processing	Stanford CS224N: NLP with Deep Learning	YouTube	10 weeks	Advanced	FREE
Ansible	Ansible — Search on Coursera	Coursera	Varies	Beginner	Paid
Prometheus	Prometheus — Search on Coursera	Coursera	Varies	Beginner	Paid
Scala	Scala — Search on Coursera	Coursera	Varies	Beginner	Paid
LangGraph	LangGraph — Search on Coursera	Coursera	Varies	Beginner	Paid
Problem Solving	Problem Solving — Search on Coursera	Coursera	Varies	Beginner	Paid
Microservices	Microservices — Search on Coursera	Coursera	Varies	Beginner	Paid
Grafana	Grafana — Search on Coursera	Coursera	Varies	Beginner	Paid
Great Expectations	Great Expectations — Search on Coursera	Coursera	Varies	Beginner	Paid
Django	Django — Search on Coursera	Coursera	Varies	Beginner	Paid
Unit Testing	Unit Testing — Search on Coursera	Coursera	Varies	Beginner	Paid
Data-Driven Decision Making	Data-Driven Decision Making — Search on Coursera	Coursera	Varies	Beginner	Paid
Snowflake	Snowflake — Search on Coursera	Coursera	Varies	Beginner	Paid
Shell Scripting	Shell Scripting — Search on Coursera	Coursera	Varies	Beginner	Paid
Project Management	Project Management — Search on Coursera	Coursera	Varies	Beginner	Paid
Jenkins	Jenkins — Search on Coursera	Coursera	Varies	Beginner	Paid
R	Machine Learning Specialization	Coursera	3 months	Beginner	Paid
R	Hands-On Machine Learning with Scikit-Learn	O'Reilly	Self-paced	Intermediate	Paid
R	Machine Learning Crash Course	Google	15 hours	Beginner	FREE
Elasticsearch	Elasticsearch — Search on Coursera	Coursera	Varies	Beginner	Paid
Vector Embeddings	Vector Embeddings — Search on Coursera	Coursera	Varies	Beginner	Paid

Skill	Course	Platform	Duration	Level	Free?
GitHub Actions	Git & GitHub Crash Course	YouTube	1 hour	Beginner	FREE
Design Patterns	Design Patterns — Search on Coursera	Coursera	Varies	Beginner	Paid
Leadership	Leadership — Search on Coursera	Coursera	Varies	Beginner	Paid
spaCy	spaCy — Search on Coursera	Coursera	Varies	Beginner	Paid
LightGBM	LightGBM — Search on Coursera	Coursera	Varies	Beginner	Paid
CrewAI	CrewAI — Search on Coursera	Coursera	Varies	Beginner	Paid
Flink	Flink — Search on Coursera	Coursera	Varies	Beginner	Paid
Recommendati on Systems	Recommendation Systems — Search on Coursera	Coursera	Varies	Beginner	Paid
Nginx	Nginx — Search on Coursera	Coursera	Varies	Beginner	Paid

■ Top Companies Hiring

CRED | Google | Infosys | Swiggy | Accenture

■ Agent Pipeline Logs

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• [Orchestrator] === Skill Gap Analysis Pipeline Started === • [Orchestrator] Candidate: Arjun Sharma | Target: Machine Learning Engineer | Mode: both • [Orchestrator] Dispatching Agent 1: JD Fetcher • [JD Fetcher] Starting JD fetch for role: 'Machine Learning Engineer' in 'India' • [JD Fetcher] Fetched 10 relevant JDs: ['Google', 'Swiggy', 'PhonePe', 'Infosys', 'Accenture', 'Walmart Labs', 'CRED', 'Paytm', 'Razorpay', 'Microsoft'] • [Orchestrator] Dispatching Agent 2: Skill Extractor • [Skill Extractor] Extracting skills from resume... • [Skill Extractor] Found 25 skills in resume: ['A/B Testing', 'Agile', 'BigQuery', 'Docker', 'ETL', 'Feature Engineering', 'Flask', 'Git', 'Google Cloud', 'JIRA', 'Jupyter Notebook', 'Linux', 'Machine Learning', 'Matplotlib', 'Model Deployment']... • [Skill Extractor] JD requires 49 unique skills across 10 JDs. • [Orchestrator] Dispatching Agent 3: Gap Analyser • [Gap Analyser] Computing skill gap analysis... • [Gap Analyser] Match: 30.6% | Matched: 15 | Gaps: 59 (34 critical/moderate, 25 optional) • [Orchestrator] Dispatching Agent 4: Recommender • [Recommender] Generating learning recommendations (mode=both, budget=10000)... • [Recommender] Generated 81 course recommendations. • [Recommender] Estimated learning timeline: 45 weeks (680 total hours). • [Orchestrator] === Pipeline Completed in 0.02s ===
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